

Sachin Bala

Sachin.bala@wmich.edu | 269-779-9757 | github.com/Sachin1bala1

May 13, 2026

Hiring Team
Amazon
Austin, TX

Dear Hiring Manager,

I am writing to apply for the Data Scientist III role at Amazon. With a strong background in statistical modeling, machine learning, and building scalable data pipelines in complex operational environments, I am excited to contribute to Amazon's data science mission and help translate large-scale business problems into actionable insights.

At Tesla, I work at exactly this intersection every day. I translate complex operational challenges into precise data requirements and build statistical models and ML pipelines using Python, R, MySQL, and JMP to optimize product performance at scale. I apply rigorous experimental design -- DOE, SPC, capability analysis, A/B frameworks -- to large manufacturing datasets, identify root causes of performance excursions, and deliver quantitative recommendations to engineering and business stakeholders. The result: two electrode production lines commissioned at 98% yield, a 10% production rate increase, and sustained quality improvements across high-volume operations.

I also built scalable automated data pipelines: a real-time Grafana monitoring system that tracks tool performance metrics across the production floor and auto-alerts on anomalies, and Python- and SQL-based analytics pipelines that surface predictive insights into tool and process health. Earlier, I built a predictive ML failure-detection model that improved manufacturing accuracy by 25%, and a Computer Vision defect detection system -- both validated with rigorous statistical methods.

Outside of work, I build AI-powered data tools: DataIQ (github.com/Sachin1bala1) is an open-source desktop IDE where an AI agent autonomously queries databases, builds analysis pipelines, and surfaces insights -- a direct reflection of my drive to make data science scalable and automated.

Amazon's culture of customer obsession, data-driven decision making, and operating at scale aligns closely with how I approach every data science problem. I am based in Austin, TX and eager to contribute to Amazon's teams here.

Thank you for your time and consideration. I look forward to the opportunity.

Sincerely,

Sachin Bala